A Data-Driven Approach for Assessing *Child Development*

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DDP - Stage 2

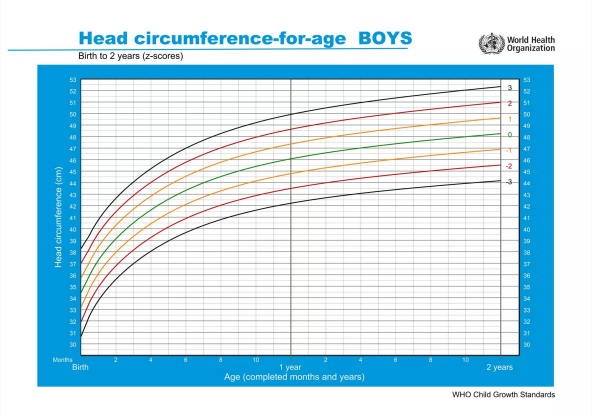
Physical Development Monitoring

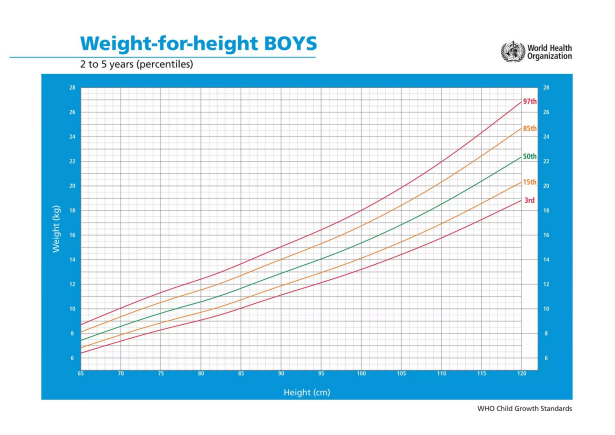
How is physical health measured in children?

- Standard Charts:- Height, Weight, and other body proportions

- And a lot more!
- Easy!

What about
MENTAL DEVELOPMENT?





MDAT

Over 150 such features!

- Conducted in India and Malawi
- Age: **0 to 6 years**
- A glimpse of data:
- Note that the data is modelled as binary: task completed or not completed

child_ids	DOBs	Sex	Age	Country	Component	ID	s1	s2	s3	s4	s5	s6	s7	s8	s9	s10	s11	s12	s13	s14	s15	s16	s17	s18	s19	s20	s21
IN-0003-BL	12/08/2018	male	1296	IN	BL	IN-0003	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
IN-0006-BL	10/08/2018	female	1301	IN	BL	IN-0006	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
IN-0007-BL	08/06/2020	male	626	IN	BL	IN-0007	1	1	1	1	-1	1	1	1	-1	-1	1	1	1	1	1	1	1	1	-1	-1	1
IN-0011-BL	01/08/2016	female	2037	IN	BL	IN-0011	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
IN-0012-BL	11/09/2021	female	172	IN	BL	IN-0012	1	1	1	1	1	-1	-1	-1	-1	-1	-1	0	0	0	0	0	0	0	0	0	0
IN-0015-BL	20/04/2018	male	1431	IN	BL	IN-0015	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
IN-0016-BL	31/01/2018	female	1485	IN	BL	IN-0016	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
IN-0019-BL	01/08/2020	female	572	IN	BL	IN-0019	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1	-1	1
IN-0021-BL	02/07/2018	female	1344	IN	BL	IN-0021	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
IN-0022-BL	26/06/2018	male	1345	IN	BL	IN-0022	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	0	0	0	1	0	1
IN-0025-BL	09/11/2020	male	490	IN	BL	IN-0025	1	1	1	1	1	1	1	1	1	1	1	1	1	-1	1	1	1	1	-1	-1	1
IN-0029-BL	15/11/2018	female	1209	IN	BL	IN-0029	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
IN-0030-BL	19/11/2020	male	498	IN	BL	IN-0030	1	1	1	1	1	1	1	1	1	-1	1	1	-1	-1	-1	-1	-1	-1	0	0	0
IN-0031-BL	06/07/2017	male	1722	IN	BL	IN-0031	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
IN-0033-BL	19/10/2016	male	1965	IN	BL	IN-0033	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
IN-0034-BL	16/03/2017	female	1825	IN	BL	IN-0034	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
IN-0035-BL	11/10/2020	female	519	IN	BL	IN-0035	1	1	1	1	1	1	1	1	1	-1	1	1	1	1	1	-1	1	1	-1	-1	1
IN-0037-BL	18/05/2017	female	1771	IN	BL	IN-0037	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
IN-0038-BL	17/05/2019	female	1040	IN	BL	IN-0038	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	0	0	0	1	0	0
IN-0040-BL	01/04/2018	male	1452	IN	BL	IN-0040	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
IN-0041-BL	24/08/2019	male	920	IN	BL	IN-0041	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	0	0	0	1	0	0
IN-0042-BL	02/07/2020	female	609	IN	BL	IN-0042	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1	-1	-1
IN-0043-BL	24/12/2019	female	805	IN	BL	IN-0043	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	-1	1
IN-0045-BL	11/09/2020	female	556	IN	BL	IN-0045	1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1	1	1	1	-1	-1	1
IN-0046-BL	08/09/2019	female	926	IN	BL	IN-0046	1	1	1	1	1	1	1	1	1	1	1	1	0	1	0	0	0	0	0	1	1
IN-0047-BL	16/05/2019	female	1041	IN	BL	IN-0047	1	1	1	1	1	1	1	1	1	1	1	1	0	0	1	0	0	0	1	0	0
IN-0048-BL	25/01/2018	female	1516	IN	BL	IN-0048	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
IN-0051-BL	18/05/2020	female	658	IN	BL	IN-0051	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	-1
IN-0052-BL	11/08/2020	female	580	IN	BL	IN-0052	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1	1

DEEP

Over 700 such features!

- Consists of **14** games
- Each game aims to capture some latent ability
- Age: **2.5 to 6 years**
- A glimpse of data:
- Mix: Ordinal, binary, continuous

childID	Age in Months	ST_Latency	ST_EndTime	ST_CORRECTCLICKS	ST_BGCLICKS	ST_TOTALCLICKS	ST_HIGHESTLEVEL	ST_ACTIVITY	ST_ACCURACY	AT_Latency	AT_EndTime	AT_CORRECTCLICKS	AT_BGCLICKS	AT_IN
IN-0003-BL	42.47835755	2.42	13.83	31	0	31	1	2.241503977	1	2.67	29.78	55	2	
IN-0006	42.54405677	2.43	14.95	8	0	8	1	0.535117057	1	3.23	28.08	8	0	
IN-0011-BL	66.82132808	3.58	14.28	15	0	15	1	1.050420168	1	2.9	29.43	31	0	
IN-0015-BL	47.5401374	2.35	6.13	14	1	15	1	2.446982055	0.933333333	1.17	29.48	30	0	
IN-0016-BL	48.7515863	0.45	11	19	4	23	1	2.090909091	0.826086957	9.12	29.48	9	4	
IN-0021-BL	43.82959115	2.17	14.97	44	0	44	1	2.939211757	1	0.13	29.72	31	0	
IN-0022-BL	44.02236869	0.3	14.88	43	0	43	1	2.889784946	1	1.35	29.83	34	6	
IN-0029-BL	39.72009182	0.22	15	49	0	49	1	3.266666667	1	0.13	29.72	37	3	
IN-0031-BL	56.5736064	3.02	14.98	44	0	44	1	2.937249666	1	1.35	29.42	22	0	
IN-0033-BL	64.22602714	2.23	14.92	49	0	49	1	3.284182306	1	2.97	29.52	46	0	
IN-0034-BL	59.94852351	1.75	15.02	48	0	48	1	3.195739015	1	0.43	29.53	48	1	
IN-0037-BL	58.51012503	2.97	15.02	49	0	49	1	3.262316911	1	8.4	29.6	36	0	
IN-0038-BL	34.1709963	2.97	15	37	0	37	1	2.466666667	1	2.35	29.38	31	1	
IN-0040-BL	47.69697028	1.87	14.73	44	0	44	1	2.987101154	1	0.4	29.7	49	1	
IN-0041-BL	30.09333403	2.45	12.97	4	0	4	1	0.308404009	1	0.1	24.35	12	3	
IN-0046-BL	30.42110808	0.28	13.1	19	1	20	1	1.526717557	0.95	0.75	25.3	14	6	
IN-0047-BL	34.19970335	0.35	14.95	33	0	33	1	2.20735786	1	3.07	29.42	3	11	
IN-0048-BL	49.80218257	2.88	14.98	41	0	41	1	2.736982644	1	0.58	29.77	51	0	
IN-0053-BL	33.01121514	1.97	13.95	4	4	8	1	0.573476703	0.5	2.95	28.28	11	0	
IN-0055-BL	41.45883789	2.18	13.93	9	0	9	1	0.646087581	1	0.47	27.32	18	0	
IN-0057-BL	35.87484563	1.35	14.7	29	3	32	1	2.176870748	0.90625	0.2	29.08	46	12	
IN-0059-BL-a	35.40906861	3.42	11.22	12	5	17	1	1.515151515	0.705882353	0.1	24.95	17	4	
IN-0061-BL	62.77998195	0.53	14.77	39	0	39	1	2.640487475	1	0.35	29.47	46	1	
IN-0063-BL	43.69394255	1.53	14.83	44	0	44	1	2.966958867	1	1.48	29.65	40	0	
IN-0064-BL	66.38952957	1.63	14.88	41	0	41	1	2.755376344	1	3.02	29.8	46	0	
IN-0067-BL	61.8930421	2.47	14.6	23	0	23	1	1.575342466	1	0.2	29.43	48	1	
IN-0068-BL	38.53713437	0.27	14.88	14	0	14	1	0.940860215	1	2.93	28.67	6	0	
IN-0069-BL	32.2250549	0.22	12.45	20	0	20	1	1.606425703	1	2.45	22.1	17	3	
IN-0070-BL	42.57083037	0.35	14.72	41	0	41	1	2.785326087	1	0.35	29.28	47	3	
IN-0071-BL	53.42087044	4.07	12.02	25	0	25	1	2.079866889	1	3.4	25.85	20	0	
IN-0074-BL	32.79291001	2.27	14.85	31	4	35	1	2.356902357	0.885714286	0.55	29.68	21	3	
IN-0075-BL	55.15215231	0.43	12.65	29	0	29	1	2.292490119	1	2.58	29.28	17	2	
IN-0076-BL	54.49879506	2.08	14.97	31	0	31	1	2.070808283	1	3.87	29.62	58	1	
IN-0077-BL	50.46216791	1.8	14.83	22	0	22	1	1.483479434	1	1.33	29.97	65	0	
IN-0078	32.42094953	0.28	0.28	1	0	1	1	3.571428571	1	0.28	29.33	26	4	
IN-0080-BL	36.42457422	2.7	11.43	19	3	22	1	1.924759405	0.863636364	2.82	27.57	10	0	
IN-0081-BL	55.48741605	3.13	14.23	14	0	14	1	0.983836964	1	2.28	29.58	34	0	
IN-0084-BL	58.84003177	2.88	14.92	36	1	37	1	2.479892761	0.972972973	0.22	29.92	51	2	
IN-0085-BL	49.04523318	1.63	14.78	36	0	36	1	2.435723951	1	1.23	29.82	58	0	

START

- Consists of **8** tasks
- **0 to 6 years:** One task - PLT (Preferential Looking Task
- **2.5 to 6 years:** All the 8 tasks
- A glimpse of the data:

Over 50 such features!

child_ids	Age	wt.PROP_TIME	wt.MEDIAN_DIST	wt.STD_DISTAI	plt.SOC_Pl	bt.soc_pref	mft.rmse	mft.jerk	ct.crossov	bpt.mean
IN-0003-B	1296	0.0822	1.49837479	-0.21956012	-0.70151	-0.94946492	-0.52037	-0.6396	-0.10029	-0.64532
IN-0006-B	1301	0.911253333	0.901356708	0.601351482	-0.84041	-0.23345801	0.650382	-0.52095	-0.29452	-0.64916
IN-0007-B	626				-0.86245					
IN-0011-B	2037	0.984733333	0.318469788	-0.46138356	0.219456	-0.23345801	-0.50665	-0.64974	0.242649	-0.61343
IN-0012-B	172				-0.84882					
IN-0015-B	1447	0.782573333	3.927318247	1.528909771	-0.20559	-2.38147874	-0.16035	0.106281	1.031523	-0.62554
IN-0016-B	1485	0.102146667	5.522597901	2.391599336	-0.0719	1.198555819	-0.47914	-0.28983	-0.66277	-0.65904
IN-0019-B	572				-0.48571					
IN-0021-B	1344	0.850106667	1.580825931	1.568311172	0.584628	0.482548906	0.94878	-0.22758	1.551602	-0.63416
IN-0022-B	1345	0.222933333	2.235951815	1.606669612		0.482548906	-0.756	-0.63663	0.444417	-0.63485
IN-0025-B	490				-0.56403					
IN-0029-B	1209	0.190093333	0.645026896	1.691780194	-0.45452	1.914562731	-0.65679	-0.66775	-0.50788	-0.62895
IN-0030-B	498				-1.75532					
IN-0031-B	1722					-0.23345801	-0.6128	-0.67328		-0.62938
IN-0033-B	1965	0.792973333	1.290137203	0.037988226	-0.41584	-0.23345801	-0.71033	-0.6491	0.87936	-0.64622
IN-0034-B	1825	0.902133333	0.213559715	-0.446480728	-0.82121	-3.09748566	-0.17009	-0.48131	-1.20748	-0.66851
IN-0035-B	519				-0.93988					

Objective

- Derive meaningful and interpretable scores using the responses received via tools

- Problems?
  - Complex Data:
    - binary, ordinal and continuous features
    - Huge Data Space (1000+ features in total)
  - Missingness in DEEP (less than 50% completion rate)
  - No labeled data available – Unsupervised?

Flow

Setting up: framework and data

Tools to measure

↓

How to derive scores?

How to validate these scores?

Tracks Age?

Against known gold standard

Against adversity measures

On Field Deployment Strategy

Item Response Theory(IRT)

- Intuition: Let's consider 4 people with unknown intelligence amount and 5 questions with unknown difficulty

Person	Q 1	Q 2	Q 3	Q 4	Q 5
A	✅	❌	❌	❌	❌
B	✅	✅	❌	❌	❌
C	✅	✅	✅	❌	❌
D	✅	✅	✅	✅	❌

Based on this **Response Matrix,**

- Can we rank the subjects based on their intelligence?
- Can we rank the Items based on their difficulty?

Item Response Theory(IRT)

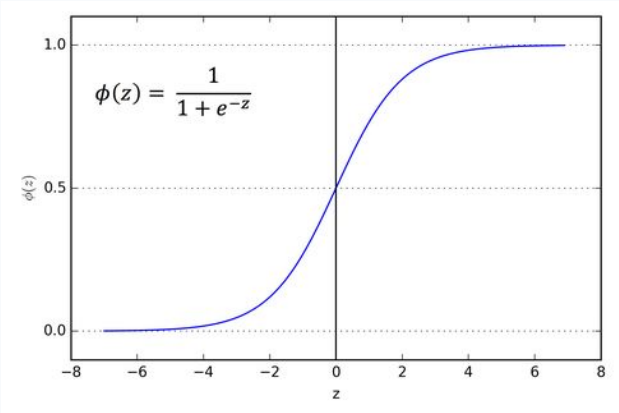
- Which parameters can be defined to model the previous problem?
- Subject Intelligence (Latent Ability) θ_i
- Item Difficulty b_j
- So, we can define: θ_i - b_j.
  - Positive: Correct
  - Negative: Incorrect
  - Equal: Can’t really say

Item Response Theory(IRT)

- So what's a good way to define a probability distribution based on this?

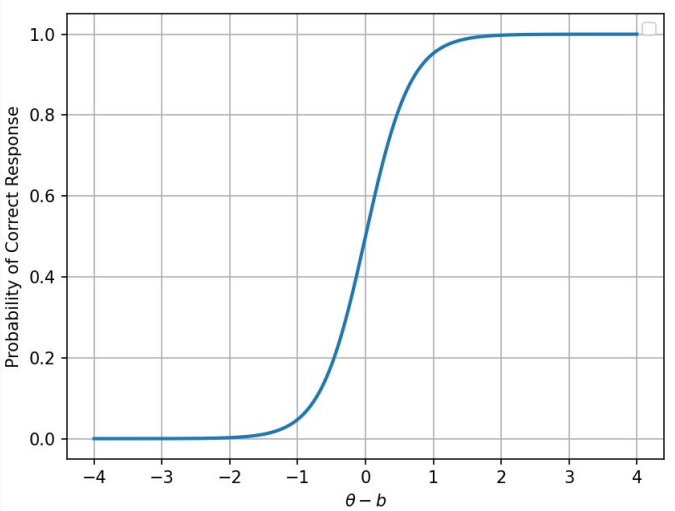
- SIGMOID!!

$$P(\text{correct} \mid \theta_i, b_j) = \frac{1}{1 + e^{-(\theta_i - b_j)}}$$



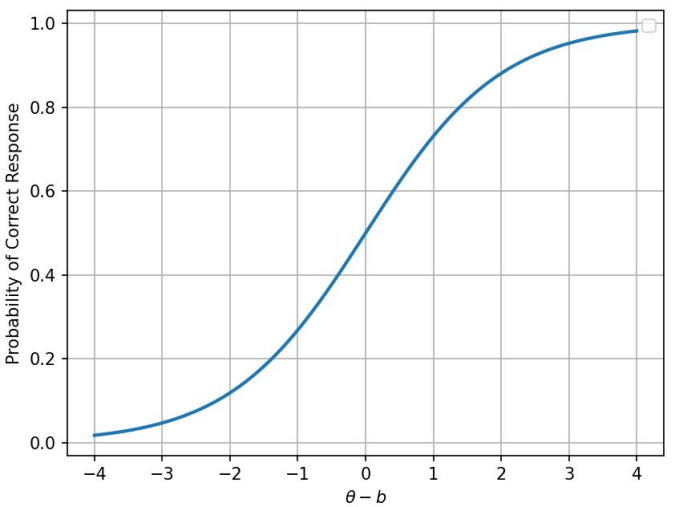
Item Response Theory(IRT)

- But the model isn't complete.
- Imagine a class full of high school students
- Question 1: $\frac{d}{dx}(x^2)$



Item Response Theory(IRT)

- But the model isn't complete.
- Imagine a class full of high school students
- Question 2: $\frac{d}{dx}\left(x^x \sin(x^x)\right)$

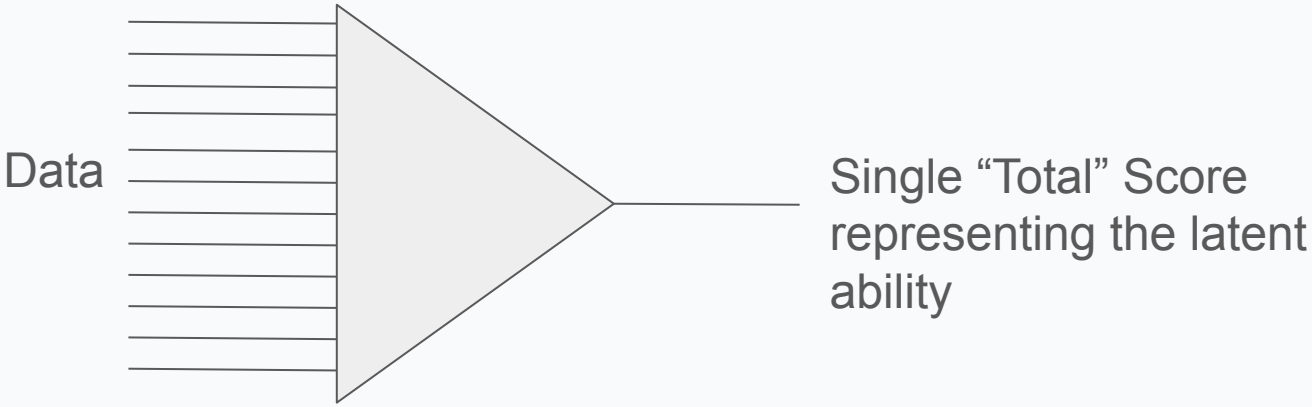


Item Response Theory(IRT)

- Which parameters can be define to model the previous problem?
- Subject Intelligence (Latent Ability) θ_i
- Item Difficulty b_j
- **Item Discrimination Parameter!** a_j
- Final Model! $$P(\text{correct} \mid \theta_i, a_j, b_j) = \frac{1}{1 + e^{-a_j(\theta_i - b_j)}}$$

How to get θ_i

- We propose **Variational Encoding** to define a mapping between the complex data and the abilities (“latent”)



Variational AutoEncoder (VAE)

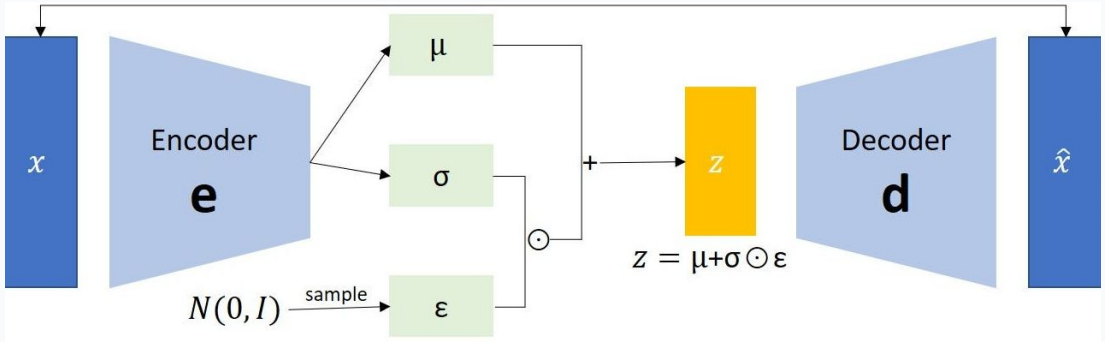
- Assuming the latent score follows normal distribution:

$$\text{Latent Score Distribution} = \text{Encoder}(\text{Responses})$$

$$q_\phi(\theta_i | X_i) = \mathcal{N}\left(\mu_\phi(X_i), \text{diag}(\sigma_\phi^2(X_i))\right)$$

- **Decoder**: (Reconstruction of responses) — Using IRT model

$$P_\psi(X_{ij} = 1 | \theta_i) = \sigma\left(a_j(\theta_i - b_j)\right) \text{ with } \psi = \{a_j, b_j\}_{j=1}^{M}$$

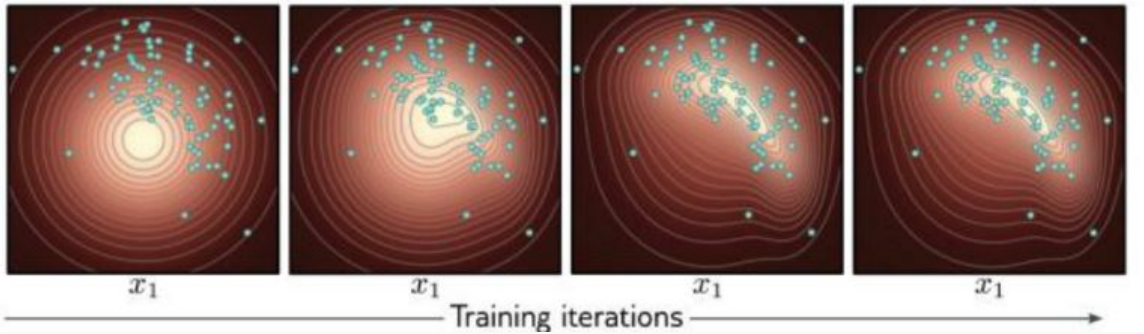


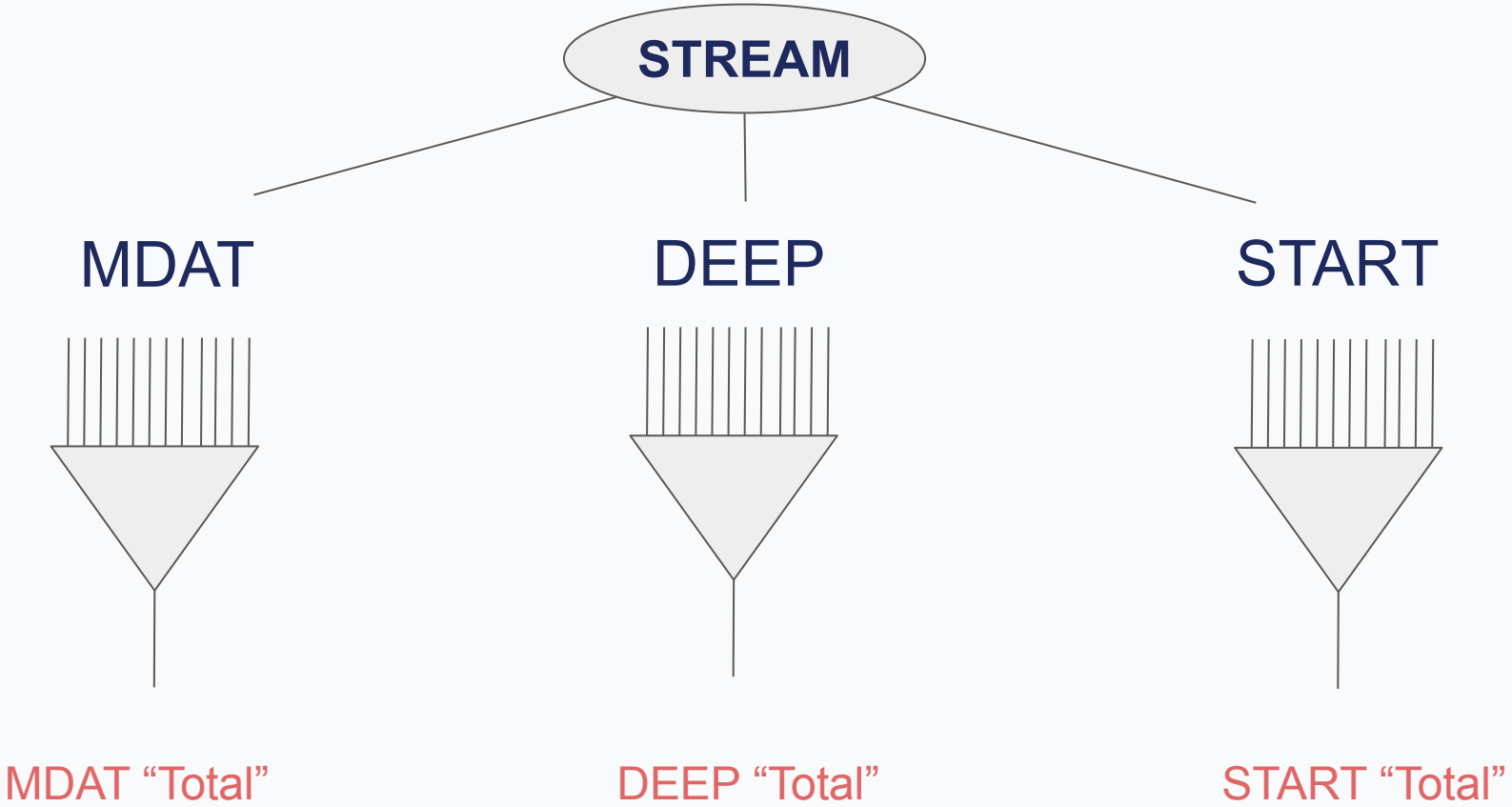
Training VAE

- **ELBO** in VAE:

$$\mathcal{L}_i(\phi,\psi)=\mathbb{E}_{\theta\sim q_\phi(\theta|X_i)}\Big[\sum_{j=1}^{M}\big(X_{ij}\log P_\psi(X_{ij}=1|\theta)+(1-X_{ij})\log(1-P_\psi(X_{ij}=1|\theta))\big)\Big]-\mathrm{KL}\big(q_\phi(\theta|X_i)\,|\,p(\theta)\big)$$

- **Objective:** $\max_{\phi,\psi}\sum_{i=1}^{N}\mathcal{L}_i(\phi,\psi)$

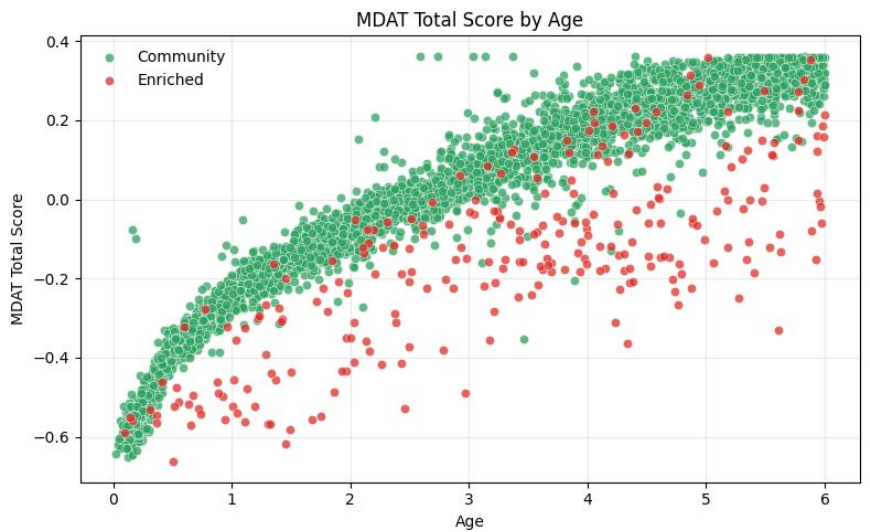



STREAM
MDAT
DEEP
START
MDAT “Total”
DEEP “Total”
START “Total”

Validity of “Total” Scores

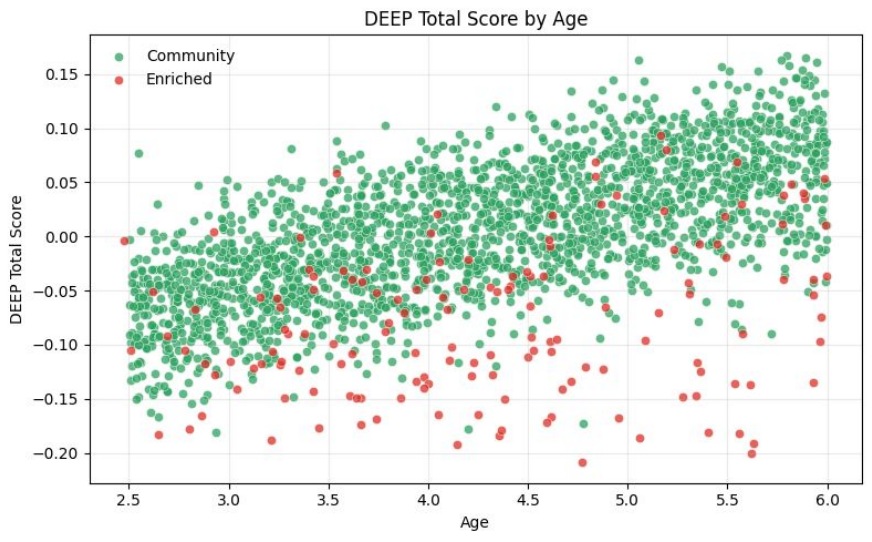
- Correlation with Age?

MDAT versus Age



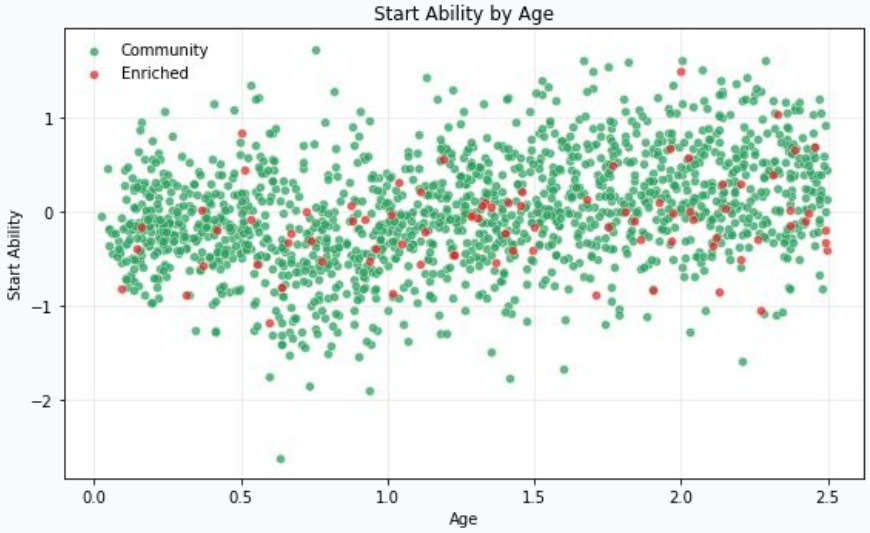
Latent Score	N	Correlation with Age	95% CI
MDAT Total	3971	**+0.901**	(+0.895, +0.907)

DEEP versus Age



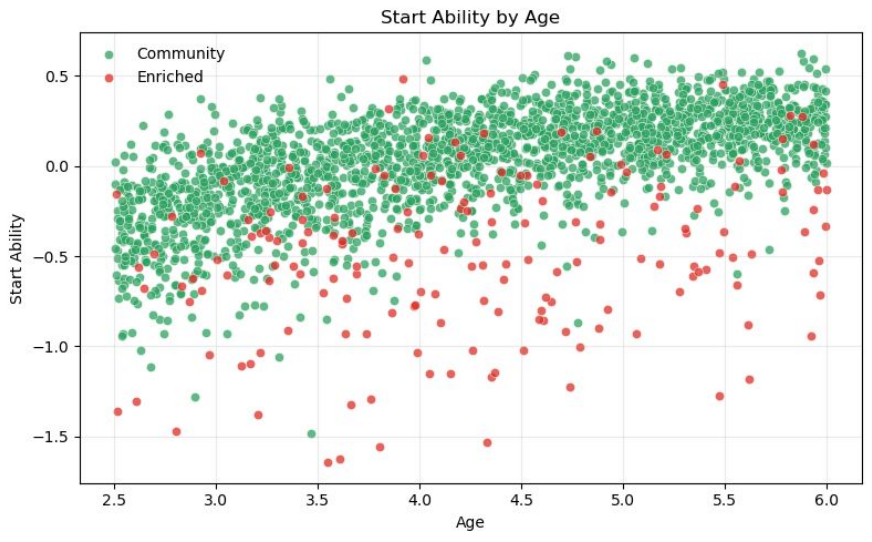
Latent Score	N	Correlation with Age	95% CI
DEEP Total	2296	**+0.616**	(+0.589, +0.641)

START (PLT) versus Age



Latent Score	N	Correlation with Age	95% CI
START Total (PLT only)	1651	**+0.330**	(+0.296, +0.381)

START (all the tasks) versus Age



Latent Score	N	Correlation with Age	95% CI
START Total (all tasks)	2342	**+0.520**	(+0.492, +0.550)

Validity of “Total” Scores

- Correlation with Age — Good!
- Performance against known Gold Standard?

GMDS (Griffith's Mental Development Scale)

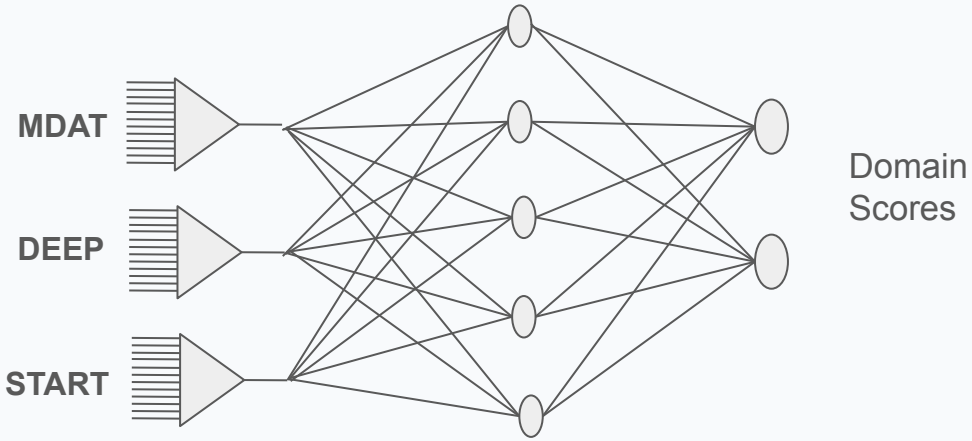
- Gold-standard tool in child development testing
- 0-6 years
- 321 items, 5 domains
  - **Foundations of learning**
  - **Language and communication**
  - **Eye and hand coordination**
  - **Personal-social-emotional**
  - **Gross motor**

Sample GMDS Results					
ChildID	Domain A	Domain B	Domain C	Domain D	Domain E
MW-0113	33	43	44	53	47
IN-1653	49	51	52	56	60
IN-1682	31	43	46	45	46

Photo of GMDS test kit

Validity of “Total” Scores

- Correlation with Age — Good!
- Performance against known Gold Standard? - Can we combine MDAT, DEEP and START latent to get domain scores?



Predicting GMDS using MDAT + DEEP + START

Domain-wise predictive performance with 5-fold cross-validation

GMDS Subscale	Mean Corr.	95% CI
FoL	**0.89**	(0.867, 0.913)
LangCom	**0.84**	(0.814, 0.866)
Eye-Hand	**0.89**	(0.862, 0.918)
Soc-Per	**0.82**	(0.800, 0.840)
Gross Motor	**0.87**	(0.844, 0.896)

Setup

Sample size: N = 560

Validation: 5-fold cross-validation

Target: GMDS subscale scores

Validity of “Total” Scores

- Correlation with Age — Good!
- Performance against known Gold Standard — Good!

- The latents work!
- Hence the method works!

Deployment Strategy

- MDAT done for 0 to 6 years old
- DEEP and START done for 2.5 to 6 years old (PLT from START for 0 to 6)

- MDAT requires a professional
- DEEP takes around 3 hours
- START also takes significant time

- Not scalable to do all three for all the children

START

Does START add anything over MDAT + DEEP?

Lower correlation with Age than MDAT and DEEP

- 0.91 for MDAT
- 0.62 for DEEP
- 0.33 (infants), 0.52 (toddlers) for START

Key finding

START does not add any signal over MDAT and DEEP.

MDAT + DEEP

GMDS Subscale	Mean Corr.
FoL	0.89
LangCom	0.85
Eye-Hand	0.89
Soc-Per	0.83
Gross Motor	0.88

MDAT + DEEP + START

GMDS Subscale	Mean Corr.
FoL	0.89
LangCom	0.84
Eye-Hand	0.89
Soc-Per	0.82
Gross Motor	0.87

Proposition

- MDAT:
  - requires a trained professional
  - is subjective
  - is less scalable
- DEEP:
  - is digital
  - Scalable

- Our experiments have shown that DEEP is better than MDAT at capturing neurodevelopment in older children!

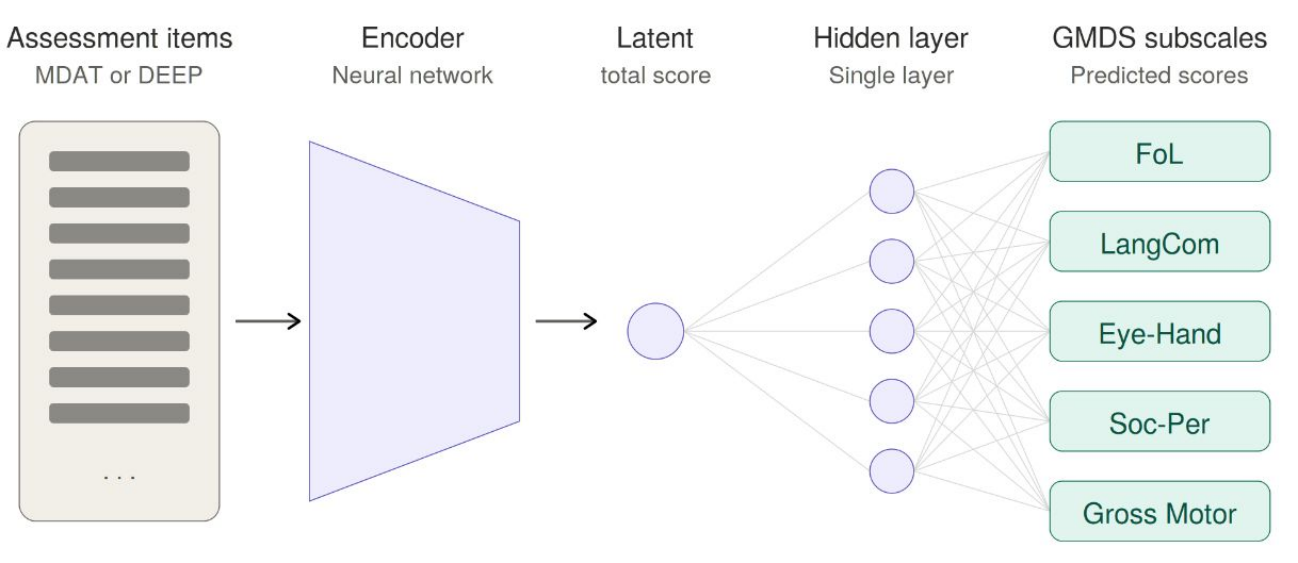
- So we propose to use MDAT for 0 to 3 year olds and DEEP for 3 to 6 year olds
  - This age split also matches GSED (Global Scales for Early Development) by WHO

How to validate the proposition?

- Correlation with Age — Good!
- Against known Gold Standard? — Using GMDS

Performance against GMDS

- We train the latents with GMDS as the target and then correlate on a held out data set.



Performance against GMDS

Held-out correlation with GMDS subscales — MDAT (N = 462) vs. DEEP (N = 484)

	MDAT (0 – 3 yrs)		DEEP (3 – 6 yrs)	
GMDS Target	**Mean Corr.**	**95% CI**	**Mean Corr.**	**95% CI**
FoL	**0.95**	(0.939, 0.961)	**0.81**	(0.787, 0.833)
LangCom	**0.95**	(0.944, 0.956)	**0.69**	(0.660, 0.720)
Eye-Hand	**0.96**	(0.949, 0.971)	**0.76**	(0.721, 0.799)
Soc-Per	**0.94**	(0.927, 0.953)	**0.58**	(0.554, 0.606)
Gross Motor	**0.96**	(0.954, 0.966)	**0.63**	(0.585, 0.675)

How to validate the proposition?

- Correlation with Age — Good!
- Against known Gold Standard — Good!
- Against anthropometric, parent and clinical experts questionnaire data? – called *convergent validity*

Convergent Variables

Anthropometric, parent-report and clinical indicators

HAZ
Height-for-Age Z-score — standardised growth indicator
Positive correlation expected

SES
Socio-Economic Status — household income, education, assets
Positive correlation expected

FCI
Family Care Indicators — home stimulation: play & caregiver activities
Positive correlation expected

MORS Warmth
Mother–Object Relations Scale, Warmth sub-scale (infants)
Positive correlation expected

MORS Invasion
MORS, Invasion sub-scale — intrusive caregiving (infants)
Negative correlation expected

CPRS Closeness
Child–Parent Relationship Scale, Closeness (toddlers)
Positive correlation expected

CPRS Conflict
CPRS, Conflict sub-scale — conflictual interactions (toddlers)
Negative correlation expected

CPAS
Child–Parent Attachment Scale (toddlers)
Positive correlation expected

RBSK
Rashtriya Bal Swasthya Karyakram — neuro-developmental risk screening
Negative correlation expected

Results

Age: 0 – 3 (MDAT)

Correlation between MDAT latent score and each convergent variable, with 95 % confidence intervals.

Feature	N	Correlation	95% CI
HAZ	1950	**+0.15**	(+0.11, +0.19)
SES	1963	**+0.03**	(-0.01, +0.08)
FCI	1967	**+0.29**	(+0.25, +0.33)
MORS Warmth	1926	**+0.17**	(+0.13, +0.21)
MORS Invasion	1926	**-0.04**	(-0.08, +0.01)
CPAS	1928	**-0.06**	(-0.11, -0.02)
RBSK	1884	**-0.41**	(-0.44, -0.37)

Results

Age: 3 – 6 (DEEP)

Correlation between DEEP latent score and each convergent variable, with 95 % confidence intervals.

Feature	N	Correlation	95% CI
HAZ	1304	**+0.15**	(+0.10, +0.20)
SES	1977	**+0.11**	(+0.07, +0.15)
FCI	1979	**+0.22**	(+0.18, +0.26)
CPRS Closeness	1874	**+0.33**	(+0.29, +0.37)
RBSK	1977	**-0.41**	(-0.45, -0.38)
CPAS	1875	**-0.02**	(-0.07, +0.02)
CPRS Conflict	1874	**-0.11**	(-0.16, -0.07)

How to validate the proposition?

- Correlation with Age — Good!
- Against known Gold Standard — Good!
- Convergent Validity — Good!

- Hence the proposition is justified!!

Future Work

Extend to other domains – multidimensional latent space!

Apply IRT-VAE pipeline beyond cognition — e.g., motor, language and social-emotional sub-domains — to validate domain-specific latents.

DEEP still takes too long!

Can we cut down certain number of games in DEEP to make it more scalable

Compare our approach (VAE) against traditional methods like PCA!

How are we doing more than just dimensionality reduction!

Larger and more diverse cohorts

Replicate findings on larger, multi-country datasets to test cross-cultural validity and refine the 0–3 / 3–6 deployment boundary.

Thank You!